

25.09.2026 items review

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September 2026

1 Context

1. How do we guarantee every UE gets served? Plain MWIS doesn't require covering every UE. A low weight UE can end up with zero scheduled slots. Two ways to address it:

- **Post-processing:** Run the solver, then check which UEs got no slot and assign each one its best remaining viable option afterward. *Simpler, cheaper.*
- **Checkpoint:** Add a coverage check inside the node removal step. *More implementation work, more robust.*¹

* Implement both behind a flag, so either can be tested during benchmarking.

2. Greedy solver via QUBO/MIS Alongside the classical Greedy MWIS baseline (Appendix A, arXiv:2301.02637), Wesley suggested formulating the same conflict graph as a QUBO for Maximum Independent Set. *Starting this now means the quantum pipeline base work is already in place once we move past the classical baseline.*

2 Items

1. Resolve fairness (A, B, or both behind a flag). **Due:** week 40
2. Implement the classical Greedy MWIS solver. **Due:** week 40
3. Build the QUBO/MIS formulation of the conflict graph. **Due:** week 40

3 Benchmarking plan

Same scenario and random seed used on every solver.

¹Can be switched on/off for a given run.

- Classical Greedy MWIS (Appendix A): the baseline.
- QUBO/MIS, solved classically (checks the QUBO formulation agrees with Greedy).
- Pulser based quantum sampling on the QPU.

Metrics: total weight (sum of spectral efficiency), coverage rate, runtime against graph size.

4 Extras

1. Prepare Paris introductory presentation.
2. Share GitHub repository access/Shared carpet.